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ChatGPT explains...

respect to each parameter θ_j:

$$\frac{\partial J(\theta)}{\partial \theta_j}$$

Step 1: Differentiate the Cost Function

Expanding $J(\theta)$:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^{m} \left[-y^{(i)} \log(h_\theta(x^{(i)})) - (1 - y^{(i)}) \log(1 - h_\theta(x^{(i)})) \right]$$

Taking the derivative with respect to θ_j:

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^{m} \left[-y^{(i)} \frac{1}{h_\theta(x^{(i)})} \frac{\partial h_\theta(x^{(i)})}{\partial \theta_j} + (1 - y^{(i)}) \frac{1}{1 - h_\theta(x^{(i)})} \frac{\partial (1 - h_\theta(x^{(i)}))}{\partial \theta_j} \right]$$

Step 2: Compute $\frac{\partial h_\theta(x)}{\partial \theta_j}$

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^{m} \left[-y^{(i)} \frac{1}{h_\theta(x^{(i)})} \frac{\partial h_\theta(x^{(i)})}{\partial \theta_j} + (1 - y^{(i)}) \frac{1}{1 - h_\theta(x^{(i)})} \frac{\partial (1 - h_\theta(x^{(i)}))}{\partial \theta_j} \right]$$

Step 2: Compute $\frac{\partial h_\theta(x)}{\partial \theta_j}$

Since $h_\theta(x)$ is the **sigmoid function**:

$$h_\theta(x) = \frac{1}{1 + e^{-\theta^T x}}$$

its derivative is:

$$\frac{\partial h_\theta(x)}{\partial \theta_j} = h_\theta(x)(1 - h_\theta(x)) x_j$$

Step 3: Substitute Back

Substituting this derivative into our previous equation:

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^{m} \left[-y^{(i)} \frac{x_j^{(i)} h_\theta(x^{(i)})(1 - h_\theta(x^{(i)}))}{h_\theta(x^{(i)})} + (1 - y^{(i)}) \frac{-x_j^{(i)} h_\theta(x^{(i)})(1 - h_\theta(x^{(i)}))}{1 - h_\theta(x^{(i)})} \right]$$

Simplifying:

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^{m} x_j^{(i)} \left(h_\theta(x^{(i)}) - y^{(i)} \right)$$

3. Gradient Descent Update Rule

Now that we have the gradient:

$$\frac{\partial J(\theta)}{\partial \theta_j} = \frac{1}{m} \sum_{i=1}^{m} x_j^{(i)} \left(h_\theta(x^{(i)}) - y^{(i)} \right)$$

we update each parameter using **Gradient Descent**:

$$\theta_j := \theta_j - \alpha \frac{1}{m} \sum_{i=1}^{m} x_j^{(i)} \left(h_\theta(x^{(i)}) - y^{(i)} \right)$$

where:

- α is the **learning rate**.
- The term $(h_\theta(x^{(i)}) - y^{(i)})$ represents the **error** in prediction.